

Cloud Computing Course

Assignment 1 – Linux Fundamentals & Shell Scripting

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Submission Instructions

1. Create a Word or PDF document and answer all theory questions using the corresponding question numbers.
2. For practical questions, include screenshots with the question number clearly mentioned.
3. Ensure screenshots clearly show the command executed and the output.
4. Submit a single PDF containing all answers and screenshots.
5. Total Marks: 50 + 5 Bonus Marks.

Section A – Linux Commands (10 Marks)

- Q1. Create the cloud_course folder structure with notes, assignments and projects folders. Attach screenshot.
- Q2. Create linux.txt and commands.txt inside notes. Copy linux.txt into projects. Attach screenshot.
- Q3. Rename commands.txt to linux_commands.txt. Attach screenshot.
- Q4. Delete the assignments directory. Attach screenshot.
- Q5. Display the current directory and hidden files. Write the commands used.

Section B – Permissions & Ownership (10 Marks)

- Q6. Create secret.txt and change permissions to 644. Attach screenshot.
- Q7. Change permissions to 755 and explain rwxr-xr-x.
- Q8. Explain the difference between chmod 777 and chmod 644.
- Q9. Write the purpose of chmod, chown and chgrp.
- Q10. Explain Owner, Group and Others.

Section C – Users & Environment Variables (10 Marks)

- Q11. Write the purpose of whoami, id and groups.
- Q12. Display your username and attach screenshot.
- Q13. Display the HOME environment variable and attach screenshot.
- Q14. Create a variable NAME with your name and display it. Attach screenshot.

Section D – Shell Scripting (10 Marks)

- Q15. Write a shell script that prints 'Welcome to Cloud Computing'.
- Q16. Write a shell script that stores your name in a variable and prints it.
- Q17. Write a shell script that accepts user input and displays 'Welcome '.

Q18. Write a shell script using an if condition to check whether a number is greater than 10.

Q19. Write a for loop that prints numbers from 1 to 5.

Section E – Linux Administration (10 Marks)

Q20. Display all running processes. Write the command used.

Q21. Write the purpose of ps, top and kill.

Q22. What is Crontab? Provide one real-world use case.

Q23. What is the purpose of ping?

Q24. Display your hostname. Write the command used.

Q25. Explain the difference between Public IP and Private IP.

Bonus Challenge (5 Marks)

Write a shell script that creates a folder named student, creates details.txt, writes your name into the file and displays the contents. Attach the script and output screenshot.